# Hailey Garcia

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Department of Mathematics, Louisiana State University, Baton Rouge, LA 70803

## Education

Ph.D. Mathematics, Louisiana State University — Fall 2025–present
*Advisors: Christin Bibby, Galen Dorpalen-Barry*

B.S. Mathematics, Louisiana State University — Fall 2022–Spring 2025

## Papers and Preprints

H. J. Garcia. *Classification of strictly resistance nonnegative graphs.* (2026). arXiv:2609.23394.

G. Agrahari, C. Bibby, S. Boros, H. J. Garcia, F. Heidercheidt, Z. Wang. *On some structural properties of graphs with non-negative resistance curvature.* (2026). arXiv:2607.13169. Code.

M. Fairchild, H. J. Garcia, J. Murphy, H. Percle. *On the non-orientable 4-genus of torus knots with braid index 5 or 6. Journal of Knot Theory and Its Ramifications* **35** (2026), No. 7, 2650001. doi:10.1142/S021821652650001X.

## Talks

On some structural properties of graphs with non-negative resistance curvature — Fall 2026
*LSU Combinatorics Seminar*

Noncontractive Function Systems — Spring 2023
*6th TX-LA Mathematics Undergraduate Conference*

On The "Kelly Sequence" — Spring 2023
*6th TX-LA Mathematics Undergraduate Conference*

## Teaching

MATH 1431, Business Calculus — Lab Leader/TA, LSU — Spring 2026
*Sections 7, 11, 19*

MATH 1431, Business Calculus — Lab Leader/TA, LSU — Fall 2025
*Sections 7, 8, 9, 11, 25, 28*

Honors Calculus 1 — Primary Instructor, Arden Cahill Academy — Fall 2023–Spring 2024

## Mentoring

Research Mentor for RTG Undergraduate — Fall 2026–present

Mentor, Directed Reading Program — Fall 2026

TA, BSM REU at Haverford College — Summer 2026

TA, 2026 LSU Undergraduate FRC — Summer 2026

Mentor, Vertically Integrated Research Courses, LSU — Fall 2025–Spring 2026

## Organisations

Organiser, LSU Informal Combinatorics Seminar — Fall 2025–present

Webmaster, LSU AMS Student Chapter — Fall 2026

Webmaster, LSU RTG — Spring 2026–present

Manager, LSU Math 3D Print Lab — Spring 2026–present

Officer, LSU AWM Student Chapter — Fall 2025–present
*President (Fall 2025–Spring 2026); Secretary (Spring 2026–present); Webmaster (Fall 2025–present)*

## Fellowships and Awards

LSU RTG Fellow — Fall 2026–present

Pasquale Porcelli Graduate Student Academic Excellence Award — Spring 2026

Robert and Betti Giles Senior Mathematics Award — Spring 2025

Peg and Tom Madden Mathematics Research Scholarship — Spring 2024

Porcelli Junior Scholarship — Spring 2023